

A COMMON BILATERAL DOUGALL FRAMEWORK FOR SUN'S CONJECTURES 3.5 AND 5.2

ABSTRACT. Let

$$W_k = \frac{\binom{2k}{k}\binom{3k}{k}\binom{6k}{3k}}{(2k+1)4096^k}, \quad g(x) = \frac{(74x+7)\Gamma(6x+1)}{(2x+1)4096^x\Gamma(3x+1)\Gamma(x+1)^3}.$$

We prove the two second-order identities and the third-order identity (3.18)–(3.20) in Sun's Conjecture 3.5, as well as the second- and third-derivative assertions in Sun's Conjecture 5.2. The first-derivative assertion in Conjecture 5.2 is not independent: after the base evaluation $\sum_{k \geq 0} (74k+7)W_k = 8$, it is exactly identity (3.17) of Conjecture 3.5. More generally, both conjectures arise from different one-parameter restrictions of the same regularized Chu–Zhang hypergeometric germ. Bilateral completion by Dougall's ${}_5H_5$ sum separates a gamma-quotient contribution from a reflected negative-index tail. The latter first appears in cubic order and is evaluated through

$$\sum_{k=0}^{\infty} \frac{H_k}{(2k+1)^2} = \frac{7\zeta(3) - \pi^2 \log 2}{4}.$$

1. STATEMENTS AND THE RELATION BETWEEN THE TWO CONJECTURES

For $r \geq 1$, write

$$H_n^{(r)} := \sum_{j=1}^n \frac{1}{j^r}, \quad H_0^{(r)} := 0, \quad H_n := H_n^{(1)}.$$

Set

$$(1) \quad W_k := \frac{\binom{2k}{k}\binom{3k}{k}\binom{6k}{3k}}{(2k+1)4096^k}.$$

Sun's Conjecture 3.5 in [2] consists of two first-order identities and three higher-order identities built from this kernel. Its underlying unweighted evaluation is Chu–Zhang's Example 60:

$$(2) \quad \sum_{k=0}^{\infty} (74k+7)W_k = 8.$$

Theorem 1.1 (Higher-order part of Sun's Conjecture 3.5). *The following identities hold:*

$$(3) \quad \sum_{k=0}^{\infty} W_k \left((74k+7)H_k^{(2)} + \frac{260(6k+1)}{(2k+1)^2} \right) = \frac{80}{3}\pi^2,$$

$$(4) \quad \sum_{k=0}^{\infty} W_k \left((74k+7)H_{2k}^{(2)} + \frac{205(6k+1)}{3(2k+1)^2} \right) = \frac{64}{9}\pi^2,$$

$$(5) \quad \sum_{k=0}^{\infty} W_k \left((74k+7)(9H_{2k}^{(3)} - H_k^{(3)}) - \frac{201(6k+1)}{(2k+1)^3} \right) = -160\zeta(3).$$

Thus equations (3.18)–(3.20) of Sun's Conjecture 3.5 are true.

2020 *Mathematics Subject Classification.* Primary 33C20; Secondary 05A19, 11M06, 33B15.

Key words and phrases. Dougall summation, bilateral hypergeometric series, harmonic numbers, gamma derivatives, Ramanujan-type series, zeta values.

Define, for $x > -1/6$,

$$(6) \quad g(x) := \frac{(74x+7)\Gamma(6x+1)}{(2x+1)4096^x\Gamma(3x+1)\Gamma(x+1)^3}.$$

Theorem 1.2 (Sun's Conjecture 5.2). *One has*

$$(7) \quad \sum_{k=0}^{\infty} g'(k) = 0,$$

$$(8) \quad \sum_{k=0}^{\infty} g''(k) = -48\pi^2,$$

$$(9) \quad \sum_{k=0}^{\infty} g'''(k) = 10752\zeta(3).$$

Hence Conjecture 5.2 of [3] holds.

The first-order overlap between the two conjectures is exact.

Proposition 1.3 (The first derivative is identity (3.17)). *For every integer $k \geq 0$,*

$$(10) \quad g(k) = (74k+7)W_k$$

and

$$(11) \quad g'(k) = 3W_k \left((74k+7)(2H_{6k} - H_{3k} - H_k - 4\log 2) + \frac{20}{2k+1} \right).$$

Consequently, in the presence of (2), equation (7) is equivalent to Sun's identity (3.17), namely

$$(12) \quad \sum_{k=0}^{\infty} W_k \left((74k+7)(2H_{6k} - H_{3k} - H_k) + \frac{20}{2k+1} \right) = 32\log 2.$$

Proof. The factorial identity

$$\frac{\Gamma(6k+1)}{\Gamma(3k+1)\Gamma(k+1)^3} = \binom{2k}{k} \binom{3k}{k} \binom{6k}{3k}$$

gives (10). Logarithmic differentiation of (6) and the formula $\psi(n+1) = H_n - \gamma$ give

$$\begin{aligned} \frac{g'(k)}{g(k)} &= \frac{74}{74k+7} - \frac{2}{2k+1} - \log 4096 + 6H_{6k} - 3H_{3k} - 3H_k \\ &= 3 \left(2H_{6k} - H_{3k} - H_k - 4\log 2 + \frac{20}{(74k+7)(2k+1)} \right), \end{aligned}$$

which is (11). Summing and using (2) yields

$$\sum_{k \geq 0} g'(k) = 3 \left(\text{left-hand side of (12)} - 32\log 2 \right).$$

□

The companion first-order identity (3.16) does not enter the equivalence in Proposition 1.3.

Remark 1.4. From second order onward the identities are related but not termwise equivalent. Repeated differentiation of g produces Bell-polynomial terms, in particular squares and cubes of the first logarithmic derivative. By contrast, the deformations used for (3)–(5) are chosen so that their first logarithmic variations vanish. The two sets of identities therefore probe different higher-order directions of one common multivariable hypergeometric identity.

2. THE COMMON HYPERGEOMETRIC GERM

For an integer n , let $(z)_n = \Gamma(z+n)/\Gamma(z)$ whenever the quotient is defined. Following Chu and Zhang [1], put

$$(13) \quad \Omega(a; b, c, d, e) := \sum_{k=0}^{\infty} (a+2k) \frac{(b)_k (c)_k (d)_k (e)_k}{(1+a-b)_k (1+a-c)_k (1+a-d)_k (1+a-e)_k}$$

and

$$(14) \quad \mathcal{G}(a; b, c, d, e) := \frac{\Gamma(1+a-b)\Gamma(1+a-c)\Gamma(1+a-d)\Gamma(1+a-e)}{\Gamma(b)\Gamma(c)\Gamma(d)\Gamma(e)} \\ \times \frac{\Gamma(b+e-a)\Gamma(c+e-a)\Gamma(d+e-a)\Gamma(1+2a-b-c-d-e)}{\Gamma(1+a-b-c)\Gamma(1+a-b-d)\Gamma(1+a-c-d)}.$$

Chu–Zhang’s Theorem 27 states, initially when $\Re(1+2a-b-c-d-e) > 0$, that

$$(15) \quad \Omega(a; b, c, d, e) - \mathcal{G}(a; b, c, d, e) = \sum_{k=0}^{\infty} \frac{(e)_{3k} (b)_k (c)_k (d)_k}{(1+a-b)_{2k} (1+a-c)_{2k} (1+a-d)_{2k}} \\ \times \frac{(1+a-b-c)_k (1+a-b-d)_k (1+a-c-d)_k}{(b+e-a)_k (c+e-a)_k (d+e-a)_k} \nu_k,$$

where

$$(16) \quad \nu_k = \frac{(a-c+2k)(a-e)}{a-c-e-k} \\ - \frac{(c+k)(e+3k)(a-e)(1+a-b-d+k)}{(1+a-d+2k)(a-b-e-k)(a-c-e-k)} \\ + \frac{(b+k)(c+k)(e+3k)(1+e+3k)(a-e)}{(1+a-b+2k)(1+a-c+2k)(1+a-d+2k)} \\ \times \frac{(1+a-b-d+k)(1+a-c-d+k)}{(a-b-e-k)(a-c-e-k)(a-d-e-k)}.$$

The factor $a-e$ in (16) motivates the regularized germ

$$(17) \quad \mathcal{D}(a; b, c, d; \varepsilon) := \frac{\Omega(a; b, c, d, a-\varepsilon) - \mathcal{G}(a; b, c, d, a-\varepsilon)}{\varepsilon}.$$

Near

$$(a, b, c, d; \varepsilon) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}; 0 \right),$$

the accelerated series (15), divided by ε , has ratio $27/64 + o(1)$ and is normally convergent. Thus (17) extends holomorphically to $\varepsilon = 0$ and defines a genuine multivariable Taylor germ.

The three deformations used below are restrictions of this same germ:

$$(18) \quad \Psi_x(t) := \mathcal{D}\left(\frac{1}{2} + t; \frac{1}{2} + (1+x)t, \frac{1}{2} + t, \frac{1}{2} + (1-x)t; t\right),$$

$$(19) \quad \Phi(t) := \mathcal{D}\left(\frac{1}{2}; \frac{1}{2} + t, \frac{1}{2} + t, \frac{1}{2} - 2t; 0\right),$$

$$(20) \quad E(u) := \mathcal{D}\left(\frac{1}{2} + 3u; \frac{1}{2} + u, \frac{1}{2} + u, \frac{1}{2} + u; 0\right).$$

The first restriction produces (3)–(4), the second produces (5), and the third is related to the translated series $\sum_{k \geq 0} g(k+u)$ and produces Conjecture 5.2.

We also need the bilateral completion

$$(21) \quad \mathcal{B}(a; b, c, d, e) := \sum_{k \in \mathbb{Z}} (a + 2k) \frac{(b)_k (c)_k (d)_k (e)_k}{(1 + a - b)_k (1 + a - c)_k (1 + a - d)_k (1 + a - e)_k}.$$

Dougall's bilateral ${}_5H_5$ summation, in this normalization, is

$$(22) \quad \mathcal{B}(a; b, c, d, e) = \frac{\Gamma(\Delta) \prod_{y \in \{b, c, d, e\}} \Gamma(1 + a - y) \Gamma(1 - y)}{\Gamma(a) \Gamma(1 - a) \prod_{\{y, z\} \subset \{b, c, d, e\}} \Gamma(1 + a - y - z)}, \quad \Delta = 1 + 2a - b - c - d - e.$$

Formula (22) follows, for example, from [4, (1.2b)] by setting $z = a/2$ and $g_y = a/2 - y$, and then multiplying by a to convert $1 + k/z$ into $a + 2k$. Euler's reflection formula gives

$$(23) \quad \mathcal{G}(a; b, c, d, e) = R(a; b, c, d, e) \mathcal{B}(a; b, c, d, e),$$

where

$$(24) \quad R(a; b, c, d, e) = \frac{\sin(\pi b) \sin(\pi c) \sin(\pi d) \sin(\pi e)}{\sin(\pi a) \sin \pi(b + e - a) \sin \pi(c + e - a) \sin \pi(d + e - a)}.$$

Let $\mathcal{N}$ be the negative-index part of (21). Replacing k by $-m - 1$ gives

$$(25) \quad \mathcal{N}(a; b, c, d, e) = -\kappa \Omega(2 - a; 1 + b - a, 1 + c - a, 1 + d - a, 1 + e - a),$$

where

$$(26) \quad \kappa = \frac{(b - a)(c - a)(d - a)(e - a)}{(1 - b)(1 - c)(1 - d)(1 - e)}.$$

Finally, Chu–Zhang's reciprocal relation [1, Theorem 5] is

$$(27) \quad \Omega(A; B, C, D, E) = \frac{A - E}{\Delta_*} \Omega(A_*; B_*, C_*, D_*, E),$$

where

$$\begin{aligned} \Delta_* &= 1 + 2A - B - C - D - E, & A_* &= 1 + 2A - B - C - D, \\ B_* &= 1 + A - B - C, & C_* &= 1 + A - B - D, & D_* &= 1 + A - C - D. \end{aligned}$$

All these identities are first used in common absolute-convergence regions and then continued meromorphically.

3. THE TWO SECOND-ORDER IDENTITIES IN CONJECTURE 3.5

Fix an auxiliary parameter x and use the first restriction (18); equivalently,

$$(28) \quad a = c = \frac{1}{2} + t, \quad b = \frac{1}{2} + (1 + x)t, \quad d = \frac{1}{2} + (1 - x)t, \quad e = \frac{1}{2}.$$

The balance is $-t$, while $a - e = t$. The accelerated expression (15) shows directly that Ψ_x is holomorphic at $t = 0$.

Since $c = a$, the negative-index tail vanishes identically. Hence $\Omega = \mathcal{B}$ on (28). Formula (22) simplifies to

$$\begin{aligned} \mathcal{B}_x(t) &:= \mathcal{B}(a; b, c, d, e) \\ &= \Gamma(-t) \frac{\Gamma(1 - xt) \Gamma(1 + xt) \Gamma(1 + t)}{\Gamma(\frac{1}{2} + t) \Gamma(\frac{1}{2} - t) \Gamma(\frac{1}{2} - xt) \Gamma(\frac{1}{2} + xt)} \\ (29) \quad &= -xt \cot(\pi t) \cot(\pi xt). \end{aligned}$$

Moreover,

$$(30) \quad R_x(t) = \frac{\cos(\pi(1+x)t) \cos(\pi(1-x)t)}{\cos^2(\pi xt)}, \quad 1 - R_x(t) = \frac{\sin^2(\pi t)}{\cos^2(\pi xt)}.$$

Because $\Omega - \mathcal{G} = (1 - R_x)\mathcal{B}_x$, we obtain the exact identity

$$(31) \quad \boxed{\Psi_x(t) = -x \frac{\sin(2\pi t)}{\sin(2\pi xt)}}.$$

At $x = 0$ the right-hand side is understood by continuity. Thus

$$(32) \quad \Psi_x(t) = -1 + \frac{2\pi^2}{3}(1-x^2)t^2 + O(t^4), \quad \Psi_x''(0) = \frac{4\pi^2}{3}(1-x^2).$$

After substituting (28) into (15), write

$$(33) \quad \Psi_x(t) = \sum_{k=0}^{\infty} A_{k,x}(t) V_{k,x}(t),$$

where

$$(34) \quad A_{k,x}(t) = \frac{(\frac{1}{2})_{3k}(\frac{1}{2} + (1+x)t)_k(\frac{1}{2} + t)_k(\frac{1}{2} + (1-x)t)_k}{(1-xt)_{2k}(1)_{2k}(1+xt)_{2k}} \\ \times \frac{(\frac{1}{2} - (1+x)t)_k(\frac{1}{2} - t)_k(\frac{1}{2} - (1-x)t)_k}{(\frac{1}{2} + xt)_k(\frac{1}{2})_k(\frac{1}{2} - xt)_k},$$

and $V_{k,x}(t)$ is the specialization of ν_k/t . At $t = 0$,

$$(35) \quad A_{k,x}(0) = Q_k := (2k+1)W_k.$$

The elementary identity

$$(36) \quad \left. \frac{d^r}{dt^r} \log(\alpha + \lambda t)_m \right|_{t=0} = (-1)^{r-1}(r-1)! \lambda^r \sum_{j=0}^{m-1} \frac{1}{(\alpha + j)^r}$$

gives

$$(37) \quad A'_{k,x}(0) = 0,$$

$$(38) \quad \left. \frac{d^2}{dt^2} \log A_{k,x}(t) \right|_{t=0} = 2(x^2 + 3)H_k^{(2)} - 6(x^2 + 4)H_{2k}^{(2)}.$$

Here we used

$$\sum_{j=0}^{k-1} \frac{1}{(j + \frac{1}{2})^2} = 4H_{2k}^{(2)} - H_k^{(2)}.$$

Direct substitution into (16) yields

$$(39) \quad V_{k,x}(0) = -\frac{74k+7}{8(2k+1)},$$

$$(40) \quad V'_{k,x}(0) = 0,$$

$$(41) \quad V''_{k,x}(0) = -\frac{5(11x^2 - 8)(6k+1)}{4(2k+1)^3}.$$

An exact rational expression for $V_{k,x}$ is recorded in Appendix A.

Differentiating (33) twice gives

$$\Psi_x''(0) = -\frac{x^2+3}{4} \sum_{k=0}^{\infty} W_k(74k+7)H_k^{(2)}$$

$$(42) \quad \begin{aligned} & + \frac{3(x^2 + 4)}{4} \sum_{k=0}^{\infty} W_k(74k + 7)H_{2k}^{(2)} \\ & - \frac{5(11x^2 - 8)}{4} \sum_{k=0}^{\infty} W_k \frac{6k + 1}{(2k + 1)^2}. \end{aligned}$$

Put

$$\begin{aligned} X &:= \sum_{k=0}^{\infty} W_k(74k + 7)H_k^{(2)}, \\ Y &:= \sum_{k=0}^{\infty} W_k(74k + 7)H_{2k}^{(2)}, \\ Z &:= \sum_{k=0}^{\infty} W_k \frac{6k + 1}{(2k + 1)^2}. \end{aligned}$$

Comparing the constant and x^2 coefficients in (32) and (42) gives

$$(43) \quad X - 3Y + 55Z = \frac{16}{3}\pi^2, \quad -3X + 12Y + 40Z = \frac{16}{3}\pi^2.$$

Solving yields

$$(44) \quad X + 260Z = \frac{80}{3}\pi^2, \quad Y + \frac{205}{3}Z = \frac{64}{9}\pi^2.$$

These are precisely (3) and (4).

4. A COMMON EULER SUM

The same elementary Euler sum occurs in both cubic calculations.

Lemma 4.1. *One has*

$$(45) \quad \sum_{k=0}^{\infty} \frac{H_k}{(2k + 1)^2} = \frac{7\zeta(3) - \pi^2 \log 2}{4}.$$

Proof. The generating function $\sum_{k \geq 0} H_k z^k = -\log(1 - z)/(1 - z)$ and $\int_0^1 x^{2k}(-\log x) dx = (2k + 1)^{-2}$ give

$$(46) \quad \begin{aligned} \sum_{k=0}^{\infty} \frac{H_k}{(2k + 1)^2} &= \int_0^1 \frac{\log x \log(1 - x^2)}{1 - x^2} dx \\ &= \frac{1}{4} \int_0^1 t^{-1/2} \frac{\log t \log(1 - t)}{1 - t} dt. \end{aligned}$$

Let $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$. The last integral equals

$$\frac{1}{4} \lim_{b \rightarrow 0^+} \partial_a \partial_b B(a, b)|_{a=1/2}.$$

As $b \rightarrow 0$,

$$\partial_a B(a, b) = -\psi_1(a) + b \left((\gamma + \psi(a))\psi_1(a) - \frac{1}{2}\psi_2(a) \right) + O(b^2).$$

Therefore

$$\lim_{b \rightarrow 0^+} \partial_a \partial_b B(a, b) = (\gamma + \psi(a))\psi_1(a) - \frac{1}{2}\psi_2(a).$$

Using

$$\psi\left(\frac{1}{2}\right) = -\gamma - 2 \log 2, \quad \psi_1\left(\frac{1}{2}\right) = \frac{\pi^2}{2}, \quad \psi_2\left(\frac{1}{2}\right) = -14\zeta(3)$$

proves (45). □

5. THE THIRD-ORDER IDENTITY IN CONJECTURE 3.5

Use the trace-free restriction (19):

$$(47) \quad a = \frac{1}{2}, \quad b = c = \frac{1}{2} + t, \quad d = \frac{1}{2} - 2t, \quad e = \frac{1}{2} - \varepsilon.$$

The balance is exactly ε . At $\varepsilon = 0$, the Pochhammer product in (15) is

$$(48) \quad A_k^{\text{tr}}(t) = \frac{(\frac{1}{2})_{3k}(\frac{1}{2} - 2t)_k(\frac{1}{2} + t)_k^2}{(1 - t)_{2k}^2(1 + 2t)_{2k}},$$

and let $V_k^{\text{tr}}(t)$ be the specialization of ν_k/ε at $\varepsilon = 0$. Then

$$(49) \quad \Phi(t) = \sum_{k=0}^{\infty} A_k^{\text{tr}}(t) V_k^{\text{tr}}(t).$$

If $\ell_k(t) = \log A_k^{\text{tr}}(t)$, formula (36) gives

$$(50) \quad \ell'_k(0) = 0, \quad \ell'''_k(0) = -12(9H_{2k}^{(3)} - H_k^{(3)}).$$

The rational factor satisfies

$$(51) \quad V_k^{\text{tr}}(0) = -\frac{74k + 7}{8(2k + 1)},$$

$$(52) \quad (V_k^{\text{tr}})'(0) = 0,$$

$$(53) \quad (V_k^{\text{tr}})'''(0) = -\frac{603(6k + 1)}{2(2k + 1)^4}.$$

Since $(A_k^{\text{tr}})'(0) = (V_k^{\text{tr}})'(0) = 0$ and $A_k^{\text{tr}}(0) = (2k + 1)W_k$, differentiation of (49) gives

$$(54) \quad \Phi'''(0) = \frac{3}{2} \sum_{k=0}^{\infty} W_k \left((74k + 7)(9H_{2k}^{(3)} - H_k^{(3)}) - \frac{201(6k + 1)}{(2k + 1)^3} \right).$$

It remains to show that $\Phi'''(0) = -240\zeta(3)$.

On (47), formula (24) becomes

$$(55) \quad R_{\text{tr}}(t, \varepsilon) = \frac{\cos^2(\pi t) \cos(2\pi t) \cos(\pi \varepsilon)}{\cos^2(\pi(t - \varepsilon)) \cos(\pi(2t + \varepsilon))}.$$

Let $D_{\text{tr}}(t, \varepsilon) = \varepsilon \mathcal{B}(t, \varepsilon)$ for the corresponding specialization of (21). Formula (22) gives

$$(56) \quad D_{\text{tr},0}(t) := D_{\text{tr}}(t, 0) = \frac{\Gamma(1 - t)^2 \Gamma(1 + 2t)}{\Gamma(\frac{1}{2}) \Gamma(\frac{1}{2} + t)^2 \Gamma(\frac{1}{2} - 2t)},$$

and

$$(57) \quad \frac{\partial_{\varepsilon} D_{\text{tr}}(t, 0)}{D_{\text{tr},0}(t)} = \delta_{\text{tr}}(t) := 2\psi(1) + \psi\left(\frac{1}{2}\right) - 2\psi\left(\frac{1}{2} - t\right) - \psi\left(\frac{1}{2} + 2t\right).$$

The negative-tail formula (25), followed by (27), gives

$$(58) \quad \mathcal{N}_{\text{tr}}(t, \varepsilon) = -C_{\text{tr}}(t) \mathcal{H}_{\text{tr}}(t, \varepsilon),$$

where

$$(59) \quad C_{\text{tr}}(t) = \frac{2t^3}{(\frac{1}{2} - t)^2 (\frac{1}{2} + 2t)}$$

and

$$(60) \quad \mathcal{H}_{\text{tr}}(t, \varepsilon) = \Omega \left(1; \frac{1}{2} - 2t, \frac{1}{2} + t, \frac{1}{2} + t, 1 - \varepsilon \right).$$

Since $\mathcal{B} = \Omega + \mathcal{N}$ and $\mathcal{G} = R\mathcal{B}$,

$$(61) \quad \Omega - \mathcal{G} = (1 - R_{\text{tr}})\mathcal{B} + C_{\text{tr}}\mathcal{H}_{\text{tr}}.$$

Put

$$(62) \quad \alpha_{\text{tr}}(t) := \partial_{\varepsilon} R_{\text{tr}}(t, 0) = \pi(\tan(2\pi t) - 2\tan(\pi t)),$$

$$(63) \quad \beta_{\text{tr}}(t) := \partial_{\varepsilon}^2 \log R_{\text{tr}}(t, 0) = \pi^2(2\sec^2(\pi t) + \sec^2(2\pi t) - 1).$$

Also, Dougall's nonterminating ${}_5F_4$ sum gives

$$(64) \quad \mathcal{H}_{\text{tr},0}(t) := \mathcal{H}_{\text{tr}}(t, 0) = \frac{\Gamma(\frac{3}{2} + 2t)\Gamma(\frac{3}{2} - t)^2\Gamma(\frac{1}{2})}{\Gamma(1 + t)^2\Gamma(1 - 2t)}.$$

Lemma 5.1 (Cancellation of the polar part). *For t near 0,*

$$(65) \quad \alpha_{\text{tr}}(t)D_{\text{tr},0}(t) = C_{\text{tr}}(t)\mathcal{H}_{\text{tr},0}(t).$$

Proof. Gamma recurrence and reflection reduce the quotient of the right-hand side by $D_{\text{tr},0}(t)$ to $\pi \tan^2(\pi t) \tan(2\pi t)$. This equals $\alpha_{\text{tr}}(t)$ by $\tan(2z) - 2\tan z = \tan^2 z \tan(2z)$. $\square$

Let

$$\mathcal{H}_{\text{tr},1}(t) := \partial_{\varepsilon} \mathcal{H}_{\text{tr}}(t, 0).$$

Expanding (61) at $\varepsilon = 0$, using $\mathcal{B} = D_{\text{tr}}/\varepsilon$ and Lemma 5.1, gives the exact master identity

$$(66) \quad \boxed{\Phi(t) = -D_{\text{tr},0}(t) \left(\alpha_{\text{tr}}(t)\delta_{\text{tr}}(t) + \frac{\alpha_{\text{tr}}(t)^2 + \beta_{\text{tr}}(t)}{2} \right) + C_{\text{tr}}(t)\mathcal{H}_{\text{tr},1}(t).}$$

Both sides are holomorphic near $t = 0$, so the identity extends there by the identity theorem.

At $t = 0$, termwise differentiation in (60) gives

$$(67) \quad \mathcal{H}_{\text{tr},1}(0) = -2 \sum_{k=0}^{\infty} \frac{H_k}{(2k+1)^2} = \frac{\pi^2 \log 2 - 7\zeta(3)}{2},$$

where the last equality is Lemma 4.1. The required expansions are

$$(68) \quad D_{\text{tr},0}(t) = \frac{1}{\pi^2} - t^2 - \frac{16\zeta(3)}{\pi^2}t^3 + O(t^4),$$

$$(69) \quad \delta_{\text{tr}}(t) = 4 \log 2 + O(t^2),$$

$$(70) \quad \alpha_{\text{tr}}(t) = 2\pi^4 t^3 + O(t^5),$$

$$(71) \quad \beta_{\text{tr}}(t) = 2\pi^2 + 6\pi^4 t^2 + O(t^4),$$

$$(72) \quad C_{\text{tr}}(t) = 16t^3 + O(t^4), \quad \mathcal{H}_{\text{tr},1}(t) = \mathcal{H}_{\text{tr},1}(0) + O(t).$$

Substitution into (66) gives

$$(73) \quad -D_{\text{tr},0} \left(\alpha_{\text{tr}}\delta_{\text{tr}} + \frac{\alpha_{\text{tr}}^2 + \beta_{\text{tr}}}{2} \right) = -1 - 2\pi^2 t^2 + (16\zeta(3) - 8\pi^2 \log 2)t^3 + O(t^4),$$

$$(74) \quad C_{\text{tr}}\mathcal{H}_{\text{tr},1} = (8\pi^2 \log 2 - 56\zeta(3))t^3 + O(t^4).$$

The logarithmic terms cancel, and hence

$$(75) \quad \Phi(t) = -1 - 2\pi^2 t^2 - 40\zeta(3)t^3 + O(t^4).$$

Thus $\Phi'''(0) = -240\zeta(3)$. Together with (54), this proves (5) and completes the proof of Theorem 1.1.

6. SUN'S CONJECTURE 5.2

Set

$$(76) \quad Q(u) := \frac{\Gamma(6u+1)}{4096^u \Gamma(3u+1) \Gamma(u+1)^3}, \quad S(u) := \sum_{k=0}^{\infty} g(k+u).$$

The multiplication formula gives

$$(77) \quad \frac{Q(k+u)}{Q(u)} = \left(\frac{27}{64}\right)^k \frac{(u+\frac{1}{6})_k (u+\frac{1}{2})_k (u+\frac{5}{6})_k}{(u+1)_k^3}.$$

Uniform Stirling estimates show that S is holomorphic near $u=0$ and that derivatives may be taken termwise. Thus

$$(78) \quad S^{(r)}(0) = \sum_{k=0}^{\infty} g^{(r)}(k).$$

It suffices to prove

$$(79) \quad S(u) = 8 - 24\pi^2 u^2 + 1792\zeta(3)u^3 + O(u^4).$$

Use the shift restriction (20):

$$(80) \quad a = \frac{1}{2} + 3u, \quad p = \frac{1}{2} + u, \quad b = c = d = p, \quad e = a - \varepsilon.$$

The balance is ε .

Lemma 6.1 (Specialized Chu–Zhang acceleration). *For parameters near $(u, \varepsilon) = (0, 0)$ and initially $\Re \varepsilon > 0$,*

$$(81) \quad \Omega(a; p, p, p, a - \varepsilon) - \mathcal{G}(a; p, p, p, a - \varepsilon) = -\frac{\varepsilon}{8} \sum_{k=0}^{\infty} A_k^{\text{sh}}(u, \varepsilon) \frac{P(k+u, \varepsilon)}{(2(k+u)+1-2\varepsilon)^3},$$

where

$$(82) \quad A_k^{\text{sh}}(u, \varepsilon) = \frac{(a-\varepsilon)_{3k} (p)_k^6}{(1+2u)_{2k}^3 (p-\varepsilon)_k^3}$$

and

$$(83) \quad \begin{aligned} P(v, \varepsilon) &= 296v^3 + 324v^2 + 102v + 7 \\ &- \varepsilon(432v^2 + 264v + 24) + \varepsilon^2(168v + 20). \end{aligned}$$

Moreover,

$$(84) \quad P(v, 0) = (2v+1)^2(74v+7), \quad A_k^{\text{sh}}(u, 0) = \frac{Q(k+u)}{Q(u)}.$$

Consequently,

$$(85) \quad E(u) = \frac{\Omega(a; p, p, p, a - \varepsilon) - \mathcal{G}(a; p, p, p, a - \varepsilon)}{\varepsilon} \Big|_{\varepsilon=0} = -\frac{S(u)}{8Q(u)}.$$

Proof. Substitution of (80) into (16) gives

$$\nu_k = -\frac{\varepsilon P(k+u, \varepsilon)}{8(2(k+u)+1-2\varepsilon)^3},$$

while the remaining Pochhammer product is (82). The factorization in (84) is immediate, and triplication of $(a)_{3k}$ together with duplication of $(1+2u)_{2k}$ gives the second identity there. The accelerated series divided by ε is normally convergent near $(0, 0)$, so evaluation at $\varepsilon = 0$ is termwise and yields (85). $\square$

On (80), equation (24) becomes

$$(86) \quad R_{\text{sh}}(u, \varepsilon) = \frac{\cos^3(\pi u) \cos \pi(3u - \varepsilon)}{\cos(3\pi u) \cos^3 \pi(u - \varepsilon)}.$$

The negative-tail formula and the reciprocal relation give

$$(87) \quad \mathcal{N}_{\text{sh}}(u, \varepsilon) = -C_{\text{sh}}(u) \mathcal{H}_{\text{sh}}(u, \varepsilon),$$

where

$$(88) \quad C_{\text{sh}}(u) = \frac{8u^3}{(\frac{1}{2} - u)^3}$$

and

$$(89) \quad \mathcal{H}_{\text{sh}}(u, \varepsilon) = \Omega \left(1; \frac{1}{2} + u, \frac{1}{2} + u, \frac{1}{2} + u, 1 - \varepsilon \right).$$

Thus

$$(90) \quad \Omega - \mathcal{G} = (1 - R_{\text{sh}}) \mathcal{B} + C_{\text{sh}} \mathcal{H}_{\text{sh}}.$$

Let $D_{\text{sh}}(u, \varepsilon) = \varepsilon \mathcal{B}(u, \varepsilon)$. Formula (22) gives

$$(91) \quad \begin{aligned} D_{\text{sh}}(u, \varepsilon) &= \frac{\Gamma(1 + 2u)^3 \Gamma(\frac{1}{2} - u)^3}{\Gamma(\frac{1}{2} + 3u) \Gamma(\frac{1}{2} - 3u) \Gamma(\frac{1}{2} + u)^3} \\ &\quad \times \frac{\Gamma(1 + \varepsilon)^2 \Gamma(\frac{1}{2} - 3u + \varepsilon)}{\Gamma(\frac{1}{2} - u + \varepsilon)^3}. \end{aligned}$$

Hence

$$(92) \quad D_{\text{sh},0}(u) := D_{\text{sh}}(u, 0) = \frac{\Gamma(1 + 2u)^3}{\Gamma(\frac{1}{2} + 3u) \Gamma(\frac{1}{2} + u)^3},$$

and

$$(93) \quad d_{\text{sh}}(u) := \frac{\partial_{\varepsilon} D_{\text{sh}}(u, 0)}{D_{\text{sh},0}(u)} = 2\psi(1) + \psi\left(\frac{1}{2} - 3u\right) - 3\psi\left(\frac{1}{2} - u\right).$$

Put

$$(94) \quad \alpha_{\text{sh}}(u) := \partial_{\varepsilon} R_{\text{sh}}(u, 0) = \pi(\tan(3\pi u) - 3\tan(\pi u)),$$

$$(95) \quad \beta_{\text{sh}}(u) := \partial_{\varepsilon}^2 \log R_{\text{sh}}(u, 0) = -\pi^2(\sec^2(3\pi u) - 3\sec^2(\pi u)).$$

Dougall's nonterminating ${}_5F_4$ sum gives

$$(96) \quad \mathcal{H}_{\text{sh},0}(u) := \mathcal{H}_{\text{sh}}(u, 0) = \frac{\Gamma(\frac{3}{2} - u)^3 \Gamma(\frac{1}{2} - 3u)}{\Gamma(1 - 2u)^3}.$$

Lemma 6.2 (Two exact cancellations). *For u near 0,*

$$(97) \quad Q(u) D_{\text{sh},0}(u) = \frac{1}{\pi^2}$$

and

$$(98) \quad \alpha_{\text{sh}}(u) D_{\text{sh},0}(u) = C_{\text{sh}}(u) \mathcal{H}_{\text{sh},0}(u).$$

Proof. The first identity follows from two applications of the duplication formula. For the second, gamma recurrence and reflection reduce $C_{\text{sh}} \mathcal{H}_{\text{sh},0} / D_{\text{sh},0}$ to $8\pi \sin^3(\pi u) / \cos(3\pi u)$, which equals $\alpha_{\text{sh}}(u)$ by $\tan(3z) - 3\tan z = 8\sin^3 z / \cos(3z)$. $\square$

Let

$$\mathcal{H}_{\text{sh},1}(u) := \partial_\varepsilon \mathcal{H}_{\text{sh}}(u, 0).$$

The defining series gives the normally convergent expression

$$(99) \quad \mathcal{H}_{\text{sh},1}(u) = -2 \sum_{k=0}^{\infty} (1+2k) \left(\frac{(\frac{1}{2}+u)_k}{(\frac{3}{2}-u)_k} \right)^3 H_k.$$

Expanding (90) at $\varepsilon = 0$ and using Lemma 6.2 yields

$$(100) \quad \boxed{S(u) = \frac{8}{\pi^2} \left(\alpha_{\text{sh}}(u) d_{\text{sh}}(u) + \frac{\alpha_{\text{sh}}(u)^2 + \beta_{\text{sh}}(u)}{2} \right) - 8Q(u) C_{\text{sh}}(u) \mathcal{H}_{\text{sh},1}(u).}$$

Both sides are holomorphic near $u = 0$, so the identity extends to a complex neighborhood of the origin.

At $u = 0$, formula (99) and Lemma 4.1 give

$$(101) \quad \mathcal{H}_{\text{sh},1}(0) = -2 \sum_{k=0}^{\infty} \frac{H_k}{(2k+1)^2} = \frac{\pi^2 \log 2 - 7\zeta(3)}{2}.$$

The elementary expansions are

$$(102) \quad \alpha_{\text{sh}}(u) = 8\pi^4 u^3 + O(u^5),$$

$$(103) \quad \beta_{\text{sh}}(u) = 2\pi^2 - 6\pi^4 u^2 + O(u^4),$$

$$(104) \quad d_{\text{sh}}(u) = 4 \log 2 + O(u^2),$$

$$(105) \quad C_{\text{sh}}(u) = 64u^3 + O(u^4), \quad Q(u) = 1 + O(u), \quad \mathcal{H}_{\text{sh},1}(u) = \mathcal{H}_{\text{sh},1}(0) + O(u).$$

Therefore

$$(106) \quad \frac{8}{\pi^2} \left(\alpha_{\text{sh}} d_{\text{sh}} + \frac{\alpha_{\text{sh}}^2 + \beta_{\text{sh}}}{2} \right) = 8 - 24\pi^2 u^2 + 256\pi^2 \log 2 u^3 + O(u^4),$$

and

$$(107) \quad -8QC_{\text{sh}}\mathcal{H}_{\text{sh},1} = (-256\pi^2 \log 2 + 1792\zeta(3))u^3 + O(u^4).$$

Again the logarithmic terms cancel. This proves (79). By (78),

$$S'(0) = 0, \quad S''(0) = -48\pi^2, \quad S'''(0) = 10752\zeta(3),$$

which proves Theorem 1.2.

Remark 6.3 (The common cubic mechanism). For the second-order deformation in Section 3, one negative-tail factor vanishes because $c = a$, so the bilateral evaluation is an exact sine quotient. For the trace-free cubic deformation in Section 5 and the shift deformation in Section 6, the negative-index prefactors are respectively

$$C_{\text{tr}}(t) = 16t^3 + O(t^4), \quad C_{\text{sh}}(u) = 64u^3 + O(u^4).$$

Thus both cubic identities see the same reflected derivative at the base point, namely $(\pi^2 \log 2 - 7\zeta(3))/2$. In each case its $\pi^2 \log 2$ contribution cancels the corresponding gamma-quotient term, leaving a pure multiple of $\zeta(3)$. This is the precise structural link between the two third-order formulas.

APPENDIX A. SPECIALIZED RATIONAL FACTORS

For completeness, we record exact rational forms underlying the derivative calculations in Sections 3 and 5. Put $n = 2k + 1$. For the second-order path,

$$(108) \quad V_{k,x}(t) = \frac{P_{n,x}(t)}{8n(n^2 - x^2t^2)(n^2 - 4x^2t^2)},$$

where

$$(109) \quad \begin{aligned} P_{n,x}(t) = & -n^4(37n - 30) + 20n^2((n - 2)x^2 + 6n - 4)t^2 \\ & + 16(4(1 - n)x^4 + (15n - 10)x^2 + 6 - 9n)t^4. \end{aligned}$$

In particular, $V_{k,x}$ is even in t .

For the trace-free third-order path,

$$(110) \quad V_k^{\text{tr}}(t) = \frac{37n^4 - 30n^3 + (-148n^3 + 120n^2)t + (384n^2 - 240n)t^2 + (-256n + 96)t^3 + 64t^4}{8(t - n)^2(4t - n)(n + 2t)}.$$

Both formulas follow by direct substitution into (16) before canceling the regularizing parameter.

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